Modern Physics Notes

Angus

2026.9.22

Abstract

This is a simple abstract for the Modern Physics notes.

Course Assessment

Class tests:
2 class tests, 10% each (special relativity + quantum mechanics).
+ 80% exam: 4 short questions (5 marks each) + 2 long questions (15 marks each).

1 Lecture 1: Galilean Relativity

1.1 Reference Frame

A reference frame is a coordinate system that specifies the position and motion of objects in space.

Definition 1.1

Frame of reference

A frame of reference is an array of rigid, massless, interpenetrable rods defining a grid over space. It is a coordinate system that allows us to specify the position of objects in space.

And this is about Newton's first law. An object at rest stays at rest, and an object in motion stays in motion with the same speed and in the same direction, unless acted upon by a net external force. Clearly, a reference frame that moves relative to an inertial reference frame at a constant speed is also an inertial reference frame.

Remark

- Inertial reference frame: a reference frame in which Newton's first law holds.
- Non-inertial reference frame: a reference frame in which Newton's first law does not hold.

Different frames can describe the same motion differently, so the choice of frame matters.

1.2 Velocity and Acceleration

In a Cartesian frame,

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k},$$

where the basis vectors are constant in time,

$$\frac{d\hat{i}}{dt} = 0, \quad \frac{d\hat{j}}{dt} = 0, \quad \frac{d\hat{k}}{dt} = 0.$$

Definition 1.2

Velocity is the rate of change of position with respect to time, and acceleration is the rate of change of velocity with respect to time.

$$\text{velocity} \quad \vec{v} = \frac{d\vec{r}}{dt},$$
$$\text{acceleration} \quad \vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2}.$$

For different theories, the definitions of velocity and acceleration remain the same, because their definitions are quite simple, but for energy and momentum, the expressions are quite different.

1.3 Time

There is only one thing to mention. In Galilean relativity, time is absolute, so velocity and acceleration depend on change in position vector; the only difference between different reference frames is different representation of the position vector. But in special relativity, time changes.

Remark

In Galilean relativity there is a master clock defining time everywhere, and time is universal. All frames read the same time instantaneously from any grid position.

1.4 Describing Motion in Different Reference Frames

1.4.1 Concepts

Different inertial reference frames give different descriptions of the same event. The position vector and velocity vector depend on the choice of the reference frame, but in Galilean relativity time is absolute, and acceleration is the same in all inertial frames.

In the standard configuration, S' moves along the $+x$ direction with constant velocity $\vec{V}$ relative to S. It does not matter which frame is called S and which is called S', as long as we are consistent.

Let frame S' move with constant velocity $\vec{V}$ relative to frame S. If the origins coincide at $t = 0$, then the position vectors are related by

$$\vec{r} = \vec{r}' + \vec{V}t, \qquad \vec{r}' = \vec{r} - \vec{V}t.$$

Differentiating with respect to time gives the velocity transformation

$$\vec{v}' = \vec{v} - \vec{V}, \qquad \vec{v} = \vec{v}' + \vec{V}.$$

Since $\vec{V}$ is constant,

$$\vec{a}' = \vec{a}.$$

In Galilean relativity, time is absolute:

$$t' = t.$$

In the standard configuration, where S' moves along the positive x direction,

$$x' = x - Vt, \quad y' = y, \quad z' = z, \quad t' = t,$$

and

$$v'_x = v_x - V, \quad v'_y = v_y, \quad v'_z = v_z.$$

Remark

In Galilean relativity: time is absolute; position and velocity depend on the reference frame; acceleration is universal in all inertial frames.

The full derivation is given again in Section 2.1.

1.4.2 Original Note

We define the position vector $\vec{r}$ and the velocity vector $\vec{v}$. They depend on the choice of the reference frame.

If we choose different reference frames, the position vector and the velocity vector will be different, related by the following equations.

$$\vec{r} = \vec{r}' + \vec{R}.$$

Remark

we assume that the position vector of particle X in frame S is $\vec{r}$ and in frame S' is $\vec{r}'$, and the position vector of the origin of frame S' relative to frame S is $\vec{R}$

Differentiating, we can get the relation of the velocity vector in different reference frames; overall it is similar to the position vector

$$\begin{aligned} \vec{v}' &= \frac{d\vec{r}'}{dt} \\ &= \frac{d\vec{r}}{dt} - \frac{d\vec{R}}{dt} \\ &= \vec{v} - \vec{V}. \end{aligned}$$

The inverse relation for velocity is given by

$$\vec{v} = \vec{v}' + \vec{V}.$$

Remember that in Galilean relativity time is absolute, $t' = t$, which allows us to build this relationship. The relation of the position vector is similar to the relation of velocity, as said, "velocity and acceleration depend on change in position vector".

2 Lecture 2: Galilean Relativity and the Michelson–Morley Experiment

> **Remark**
> Relativity is the relationship of one reference frame to another.

2.1 Review of Galilean Transformation

2.1.1 Relations

Let frame S' move with constant velocity $\vec{V}$ relative to frame S. For a particle, denote its position by $\vec{r}$ in S and $\vec{r}'$ in S'. If the origins coincide at $t = 0$, then

$$\vec{R} = \vec{V}t,$$

and

$$\vec{r} = \vec{r}' + \vec{R} = \vec{r}' + \vec{V}t.$$

Therefore

$$\vec{r}' = \vec{r} - \vec{V}t.$$

Differentiating with respect to time,

$$\vec{v}' = \frac{d\vec{r}'}{dt} = \vec{v} - \vec{V},$$

or

$$\vec{v}' = \vec{v} - \vec{V}.$$

The inverse transformation is

$$\vec{v} = \vec{v}' + \vec{V}.$$

Since $\vec{V}$ is constant,

$$\frac{d\vec{V}}{dt} = 0,$$

so

$$\vec{a}' = \vec{a}.$$

If S' is non-inertial, i.e. $\vec{V} = \vec{V}(t)$, then

$$\vec{a}' = \vec{a} - \vec{A}, \qquad \vec{A} = \frac{d\vec{V}}{dt}.$$

And finally,

> **Remark**
> In Galilean relativity, $t' = t$; time is absolute.

2.1.2 Original Note and Example

A frame S' moves with constant velocity $\vec{V}$ relative to another frame S.
According to Galilean relativity,

$$\begin{aligned}
\vec{R} &= \vec{V}t, \\
\vec{r}' &= \vec{r} - \vec{R}, \\
\dot{\vec{R}} &= \vec{V} = \text{const}, \\
\vec{r}' &= \vec{r} - \vec{R} = \vec{r} - \vec{V}t.
\end{aligned}$$

The position transformation does not change time.

$$\begin{aligned}
t' &= t, \\
\vec{v}' &= \vec{v} - \vec{V}, \\
\vec{a}' &= \vec{a} - \vec{A}.
\end{aligned}$$

Where $\vec{a}'$ is the acceleration in frame S', $\vec{a}$ is the acceleration in frame S, and $\vec{A} = d\vec{V}/dt$ is the acceleration of frame S' relative to frame S.

In inertial frames, $\vec{A} = 0$, so $\vec{a}' = \vec{a}$.

> **Remark**
>
> In Galilean relativity, acceleration is universal.

For example, on a ship moving with constant velocity, a ball dropped from the crow's nest falls straight down relative to the ship, even though the ship is moving relative to the ground.

2.2 Introduction to the Michelson-Morley Experiment

- At the beginning of the 1800s, Young's interference experiment demonstrated the wave nature of light.
- A wave requires a medium for propagation, leading to the hypothesis of the luminiferous aether.

Galilean relativity works well for mechanics, but it is not compatible with Maxwell's electromagnetism. Maxwell's equations give a fixed speed of light c, and experiments always measured the same speed of light, independent of the motion of the Earth.

At that time, physicists assumed light needed a medium called the luminiferous aether. The Michelson–Morley experiment was designed to detect the Earth's motion relative to this aether.

2.2.1 Hand-Clapping Experiment

In this experiment, two people stand a certain distance apart and clap their hands when they hear another's clap.

- case 1: platform is stationary relative to the ground.

- case 2: platform is moving relative to the ground.
- case 3: the line joining the two people is perpendicular to the direction of motion.

For case 1,

$$t = \frac{2L}{v_s}.$$

For case 2,

$$t = \frac{2Lv_s}{v_s^2 - v_p^2}.$$

For case 3,

$$t = \frac{2L}{\sqrt{v_s^2 - v_p^2}}.$$

Clearly,

$$t_1 \geq t_2.$$

As a scientific check, when $v_p = 0$, we recover

$$t = \frac{2L}{v_{\text{sound}}}.$$

2.2.2 Michelson–Morley Experiment

The main idea is not to measure T_1 and T_2 individually, we use a beam splitter, look at the interference pattern.

- If the two travel times are different, the waves are out of phase, and the interference pattern will shift.
- If no shift takes place, it means the two travel times are equal. No aether was detected.

The total travel times for the two arms of the interferometer are given by

$$T_1 = t_f + t_b = \frac{2Lc}{c^2 - v^2} \qquad \text{(parallel arm)},$$
$$T_2 = \frac{2L}{\sqrt{c^2 - v^2}} = T_1\sqrt{1 - \frac{v^2}{c^2}} \quad \text{(perpendicular arm)}.$$

For the parallel arm, the light travels with speed $c + v$ in one direction and $c - v$ in the other, so

$$T_1 = \frac{L}{c+v} + \frac{L}{c-v} = \frac{2Lc}{c^2 - v^2}.$$

For the perpendicular arm, the light travels along the hypotenuse of a right triangle, so

$$\left(\frac{cT_2}{2}\right)^2 = L^2 + \left(\frac{vT_2}{2}\right)^2,$$

which gives

$$T_2 = \frac{2L}{\sqrt{c^2 - v^2}}.$$

2.2.3 Conclusion: Lorentz-FitzGerald Contraction

Lorentz and FitzGerald proposed that objects contract along the direction of motion, by the factor $1/\gamma$, in order to explain the null result.

$$1/\gamma = \sqrt{1 - \frac{v^2}{c^2}}.$$

This was mathematically close to the correct transformation, but for the wrong physical reason. Einstein later explained the null result by postulating that the speed of light is the same in all inertial frames.

3 Lecture 3: Foundations of Special Relativity

3.1 Natural Units

Definition 3.1

An electron accelerated through a potential difference of 1 V gains kinetic energy 1 eV.

3.2 Van de Graaff Accelerator

In 1909, Bücherer used a Van de Graaff generator to accelerate electrons and measured their speed and kinetic energy. The experiment showed that the speed approaches the speed of light and that the classical formula fails at high energies,

$$T = \frac{1}{2}mv^2.$$

Plotting the speed and kinetic energy of electrons from experimental data shows that the speed of electrons approaches a significant fraction of the speed of light.

Remark

The speed of light is always c. This is an experimental fact.

3.3 The Postulates of Special Relativity

3.3.1 Postulate 1

Postulate 1: The laws of physics are the same in all inertial frames. All inertial (non-accelerating) frames are ”equivalent”. It is the same as the familiar principle of Galilean relativity.

3.3.2 Postulate 2

Postulate 2: The speed of light in a vacuum is the same value c for any inertial frame.

Remark

The Michelson–Morley experiment supported this postulate; it was unable to detect any difference in the speed of light due to the motion of the Earth through the supposed aether. There is a limiting speed for any object in any frame.

3.3.3 Units of Measurement

Definition 3.2

- Speed of light: $c = 299\,792\,458\ \mathrm{m\,s^{-1}}$.
- Second: defined by the hyperfine transition frequency of Cs-133.
- Meter: defined by the distance light travels in vacuum in $1/299\,792\,458$ seconds.

3.4 Transverse Rods

Consider rod A moving perpendicular to rod B of the same proper length. Suppose the transverse length changed by a factor $f(v)$, so that A would measure B's transverse length as $f(v)L$, and B would measure A's transverse length as $f(v)L$.

By symmetry,

$$f(v) = f(-v).$$

But the transverse direction has no preferred orientation, so no such factor can appear.

Therefore $f(v) = 1$, and the length of a rod moving perpendicular to the direction of motion remains unchanged. A more rigorous proof uses two metre sticks and pens. If the moving stick were shorter in both frames, contradictions arise; hence transverse lengths must be unchanged.

Remark

Perpendicular length remains unchanged.

3.5 The Role of Observers

Measurement should not disturb the system being measured.

3.6 The Frame of Reference in Special Relativity

- relativity of simultaneity
- time dilation
- length contraction

We need to define time at different points in a consistent manner for all observers.

4 Lecture 4: Introduction to Special Relativity

4.1 Time in Special Relativity

In special relativity, time is not absolute and can vary depending on the relative motion of observers. Every coordinate has its own time. We can use four-vector notation (x, y, z, ct) to represent spacetime events.

4.2 Synchronising Clocks

A light source sends photons from the middle of the rod. In the rod's rest frame, both photons reach the ends at the same time. This allows observers to synchronise clocks at both ends. Repeating this at different locations generates a grid of synchronised clocks.

As a result, this defines simultaneity in that frame.

4.3 Events

Definition 4.1

An event is where and when something happens.

A reference frame provides the coordinate system and clocks used to assign position and time to events.

Relativity deals with events in (x, y, z, t).

4.4 Proper Time

Definition 4.2

The time interval between two events within the same spatial coordinates is called the proper time.

All other time intervals are improper time intervals.

If, in a given reference frame, the two events occur at different spatial positions, then the time interval measured in that frame is improper.

4.5 Time Dilation

Definition 4.3

Time dilation

$$\tau = \gamma\tau_0, \quad \text{where } \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}.$$

- proper time τ_0: the time interval between two events occurring at the same spatial location.
- dilated time τ: the time interval measured in a frame where the clock is moving.
- Lorentz factor $\gamma = 1/\sqrt{1 - v^2/c^2}$.

- scenario 1: Observer and clock are stationary relative to each other.
- scenario 2: Observer and clock are moving relative to each other.

$$\begin{aligned} &\tau_0: \text{ proper time} && \text{(scenario 1)},\\ &\tau = \gamma\tau_0 && \text{(scenario 2)},\\ &\gamma = \frac{1}{\sqrt{1-v^2/c^2}} && \text{(Lorentz factor)}. \end{aligned}$$

4.5.1 Light Bouncing Between Mirrors

A platform carrying two mirrors, one at the top and one at the bottom, moves relative to the ground frame S. We have three events.

- Event 1: Light leaves the source.
- Event 2: Light reaches the top mirror.
- Event 3: Light reaches the bottom mirror.

For the platform frame S', the coordinates of the events are as follows.

Event	x	y	z	t
1	0	0	0	0
2	0	l_0	0	l_0/c
3	0	0	0	$2l_0/c$.

The time interval between events 1 and 3 in S' is $2l_0/c$. It is proper time because the position of the clock (light source) does not change in S'.

$$\Delta t'_{13} = \tau_0 = \frac{2l_0}{c}.$$

When observed from the ground frame S, the light travels a longer diagonal path due to the motion of the platform.

The length of the diagonal path traveled by the light is l_{eff}. The three events are listed as follows in the ground frame S.

Event	x	y	z	t
1	0	0	0	0
2	vl_{eff}/c	l_0	0	l_{eff}/c
3	$2vl_{\text{eff}}/c$	0	0	$2l_{\text{eff}}/c$.

The y-coordinate is unchanged because the motion is along x.

The time interval between events 1 and 3 in S is $2l_{\text{eff}}/c$; it is improper time because the clock (light source) is moving in S.

$$\Delta t_{13} = \frac{2l_{\text{eff}}}{c} = \tau.$$

Because the light travels in the long diagonal path in the ground frame S, we can use the Pythagorean theorem to relate l_{eff} and l_0, and the distance traveled due to the platform's motion.

$$l_{\text{eff}}^2 = l_0^2 + \left(v\frac{l_{\text{eff}}}{c}\right)^2.$$

Solving for l_{eff},

$$l_{\text{eff}} = \frac{l_0}{\sqrt{1 - v^2/c^2}} = \gamma l_0.$$

And also we can relate the time intervals in the two frames, using the Pythagorean theorem.

$$l_0{}^2 + \left(\frac{v\tau}{2}\right)^2 = \left(\frac{c\tau}{2}\right)^2.$$

Solving for τ,

$$\begin{aligned}
\tau &= \frac{2l_0}{\sqrt{c^2 - v^2}} = \frac{2l_0}{c\sqrt{1 - v^2/c^2}}, \\
\tau &= \gamma \frac{2l_0}{c}, \\
\tau_0 &= \frac{2l_0}{c}, \\
\tau &= \gamma \tau_0.
\end{aligned}$$

4.6 The Lorentz Factor γ

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}.$$

- γ is the Lorentz factor. Time intervals are multiplied by γ, lengths along the direction of motion are divided by γ, and relativistic momentum and energy involve γ.
- By definition, $\gamma \geq 1$.
- γ becomes very large as v approaches c, indicating significant relativistic effects.
- When $\gamma = 1$, the relative velocity $v = 0$, indicating no relativistic effects.

4.6.1 On Time Dilation

- Proper time is the time interval between two events occurring at the same spatial location.
- Time dilation occurs when observing a moving clock, which appears to run slower than a stationary clock.
- Proper time: τ_0. Dilated time: $\tau = \gamma \tau_0$.

If A and B move relative to each other, each observes the other's clock running slower. This is not a contradiction; it depends on which frame is used.

4.6.2 Experimental Proof

We use the muon lifetime as an experimental proof of time dilation.

Muons are produced in the upper atmosphere and travel towards the Earth's surface. Due to their short lifetime, without time dilation, very few muons would reach the Earth's surface.

However, because of time dilation, moving muons experience a longer lifetime from the perspective of an observer on Earth, allowing many of them to reach the surface.

Use τ_0 as the mean lifetime of the muon at rest; for a moving muon, the effective mean lifetime is

$$\tau = \gamma\tau_0.$$

And for a radioactive sample,

$$-\frac{dN}{dt} \propto N,$$
$$N(t) = N_0 e^{-t/\tau_{\text{mean}}}.$$

Then setting

$$N(t_{1/2}) = N_0/2$$

gives

$$t_{1/2} = \tau_{\text{mean}} \ln 2.$$

For a moving muon, the classically expected surviving fraction is $P = 0.05$, but experimentally $P = 0.72 \pm 0.03$. Special relativity predicts $P = 0.75$.

Remark

The discrepancy between the classical expectation and the experimental observation is explained by time dilation, which increases the effective lifetime of moving muons as observed from the Earth's frame.

5 Lecture 5: Space and time in Special Relativity

5.1 Lorentz Contraction

5.1.1 Definition

Definition 5.1

Lorentz contraction

$$l = l_0 \frac{1}{\gamma}, \quad \text{where } \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}.$$

1. proper length l_0: event time coordinates are the same when measuring the length of an object, length in the system at rest.

2. improper length l: length when viewed moving at speed v.

Example

Length Contraction of a Muon

In the muon frame, the distance between the muon and earth is Lorentz contracted compared to the distance measured in the Earth's frame.

5.1.2 Thought experiment: the moving train

The train is moving at a velocity v relative to an observer on the platform.

Proper length

Observer in the middle of the platform sends two flashes of light (same technique used to synchronise clocks)

- the light reaches the first end of the platform

$$(x_1 = 0, t_1 = T).$$

- the light reaches the far end of the platform

$$(x_2 = 2cT, t_2 = T).$$

Length of the platform is $l_0 = 2cT$, because the events occur at the same time in the platform frame, $t_1 = t_2$, so l_0 is the proper length.

In the Platform Frame

- The train passes at one end of the platform: $(0, t_1)$.
- The train passes at the far end of the platform: (l, t_2).

The time interval is

$$\Delta t = t_2 - t_1 = \tau.$$

The space coordinates are different

$x_1' \neq x_2' \quad \Rightarrow \quad \tau$ is not a proper time, the events are not seen at the same coordinate.

The observer on the platform confirms that

$$l_0 = v\tau.$$

In the train driver frame of reference

The driver is not moving while the platform is passing at a speed v.

- Event 1, corresponding to the near end of the platform: $(x_1' = 0, t_1')$.
- Event 2, corresponding to the far end of the platform: $(x_2' = 0, t_2')$.

The time interval is

$$\Delta t = t'_2 - t'_1 = \tau_0.$$

The space coordinates are the same

$$x'_1 = x'_2 \quad \Rightarrow \quad \tau_0 \text{ is a proper time, the events are seen at the same coordinate.}$$

The deduced length of the platform is

$$\Delta x' = l = v\tau_0.$$

Since $t'_1 \neq t'_2$, the length l is an improper length: the events are not seen at the same time in the train's frame.

Length contraction

The two lengths determined above are

$$\begin{aligned} l_0 &= v\tau, \\ l &= v\tau_0. \end{aligned}$$

The ratio between the lengths observed by the two observers is

$$\begin{aligned} \frac{l}{l_0} &= \frac{v\tau_0}{v\tau}, \\ \tau &= \gamma\tau_0, \\ l &= \frac{l_0}{\gamma}. \end{aligned}$$

And length contraction takes place. This is equivalent to the time dilation effect.

5.1.3 The twin paradox

Twin A on earth remains stationary, while Twin B travels on a spaceship at a high velocity.

1. From A's perspective, Twin B's clock runs slower due to time dilation. As a result, when Twin B returns to Earth, they will have aged less than Twin A, despite both experiencing the passage of time according to their own frames of reference.

2. In B's frame, it is Twin A who appears to be moving, and thus Twin A's clock runs slower from Twin B's perspective. However, the asymmetry arises because Twin B undergoes acceleration and deceleration during the journey, changing inertial frames, while Twin A remains in a single inertial frame. This explains why Twin B ages less upon returning to Earth.

The error lies in the fact that B doesn't remain in the same inertial frame throughout the journey, while A does. B accelerates while changing direction, which breaks the symmetry and results in B aging less than A upon reunion.

5.2 Moving Synchronised Clocks

Consider two clocks that are synchronised in their own rest frame and are moving with a constant velocity relative to an observer. Let frame S move with respect to the page, and let frame S' be the page.

In the rest frame of the clocks, the two clocks are synchronised, and their separation is Δx. In the page frame S', the two clocks are moving with speed v, and the distance between them is contracted.

$$AM = BM = \frac{\Delta x}{2\gamma}.$$

The light signals emitted from the midpoint reach the two clocks at

$$\begin{aligned} ct'_A &= \frac{\Delta x}{2\gamma} - vt'_A \quad \Longrightarrow \quad t'_A = \frac{\Delta x}{2\gamma(c+v)}, \\ ct'_B &= \frac{\Delta x}{2\gamma} + vt'_B \quad \Longrightarrow \quad t'_B = \frac{\Delta x}{2\gamma(c-v)}. \end{aligned}$$

Therefore,

$$t'_A \neq t'_B \quad \Longrightarrow \quad \Delta t' = t'_B - t'_A = \frac{\gamma v \Delta x}{c^2}.$$

Since $t'_A < t'_B$, the leading clock lags behind the trailing clock.

Definition 5.2

Leading clock lags (clock lag)

If two clocks are synchronised in their rest frame and move with velocity v relative to an observer, the leading clock lags behind the trailing clock by

$$\Delta t = \frac{\gamma v L}{c^2},$$

where $L = \Delta x$ is the proper distance between the two clocks.

Remark

The clocks are not synchronised when viewed from a frame moving with a different speed. This is a direct consequence of the relativity of simultaneity.

5.2.1 The Rod-Shed Paradox

The rod-shed paradox involves a rod moving at a high velocity towards a shed. The length of shed is L_0, the length of the rod is l_0.

$$\begin{aligned} l_0 &= 2L_0, \\ v &= \frac{\sqrt{3}}{2}c, \\ \gamma &= \frac{1}{\sqrt{1 - v^2/c^2}} = 2. \end{aligned}$$

Shed frame

In the shed frame, the rod can be fully contained within the shed due to length contraction.

$$l = \frac{l_0}{\gamma} = \frac{2L_0}{2} = L_0.$$

Rod frame

In the rod's frame, the shed is length contracted and appears too short to contain the rod.

$$l = l_0,$$
$$L = \frac{L_0}{\gamma} = \frac{L_0}{2}.$$

In the rod's frame, clock D leads clock C.

1. C reaches A $\rightarrow$ door C closes.
2. D reaches B $\rightarrow$ door D closes.
3. Door D closed after door C: "The leading clock lags".

The synchronised clocks in the rod's frame are not simultaneous with those in the shed's frame due to the relativity of simultaneity.

> **Remark**
>
> The relativity of simultaneity implies that events that are simultaneous in one frame are not necessarily simultaneous in another frame moving relative to the first.

5.2.2 Transverse and Parallel Motion

From the perspective of leading clock lag.

1. In the transverse motion, the proper length can consistently be measured: both ends can always be measured at the same time, because the motion is perpendicular to the direction of measurement.
2. In the parallel motion, however, the proper length cannot be measured simultaneously in the moving frame because the motion is along the direction of measurement, since the leading clock lags, leading to length contraction.

> **Remark**
>
> In transverse motion, both ends can be measured simultaneously; in parallel motion, the leading clock lags, so the ends cannot be measured simultaneously, leading to length contraction.

5.3 Summary

Time dilation in one frame is seen as length contraction in another; they are the same effect.

- time dilation: "a moving clock runs slow"

$$\tau = \gamma\tau_0.$$

- space contraction: "a moving stick is shorter"

$$l = \frac{l_0}{\gamma}.$$

- synchronisation: "a leading clock lags"

$$\Delta t = \frac{\gamma v L}{c^2}.$$

6 Lecture 6: Lorentz Transformation

6.1 Frame of Reference

Frame of reference S' moving with velocity v relative to frame S along the x-axis.

$$\begin{aligned}
\Delta x &= A\Delta x' + B\Delta t', \\
\Delta t &= C\Delta t' + D\Delta x', \\
\Delta y &= \Delta y', \\
\Delta z &= \Delta z'.
\end{aligned}$$

We assume that the transformations are linear and that the A, B, C, and D are four constants to be determined.

6.2 Derivation of Lorentz Transformation

- S' moves with velocity v with respect to S.
- S moves with -v with respect to S'.
- The laws of physics are the same in two reference systems. $\Longrightarrow$ If there is time dilation in one frame, there must be a corresponding time dilation in the other frame.
- a light pulse has speed c is S and S'.

6.2.1 S' Moving with Velocity v Relative to S

Consider a rod at rest in S',

$$\Delta x' = 0.$$

According to the Lorentz transformation,

$$\begin{aligned} \Delta x' &= 0, \\ \Delta x &= B\Delta t', \\ \Delta t &= C\Delta t'. \end{aligned}$$

Dividing Δx by Δt,

$$\frac{\Delta x}{\Delta t} = \frac{B\Delta t'}{C\Delta t'} = \frac{B}{C} = v \implies B = vC.$$

So

$$\begin{aligned} \Delta x &= A\Delta x' + vC\Delta t', \\ \Delta t &= C\Delta t' + D\Delta x'. \end{aligned}$$

6.2.2 S Moving with Velocity -v Relative to S'

Consider the space interval in S moving with velocity -v with respect to S'.

$$\begin{aligned} \Delta x &= 0, \\ 0 &= A\Delta x' + B\Delta t'. \end{aligned}$$

So

$$\begin{aligned} \Delta x' &= \frac{-vC}{A}\Delta t', \\ \Delta t' &= -v = -\frac{vC}{A}. \end{aligned}$$

Then

$$v = \frac{vC}{A} \implies A = C.$$

So we have

$$\begin{aligned} \Delta x &= A\Delta x' + vA\Delta t', \\ \Delta t &= A\Delta t' + D\Delta x'. \end{aligned}$$

6.2.3 Time Interval for Clocks at Rest in Different Frames

Consider a clock A at rest in S' $\implies \Delta x' = 0$.

- Event 1 tick 1 x'_1,
- Event 2 tick 2 $x'_2 = x'_1$, $\Delta x' = 0$.

The time interval between two ticks is $\Delta t'$,

$$\Delta t_A = A\Delta t'_A.$$

Clock B at rest in S $\implies \Delta x = 0$.

- Event 1 tick 1 x_1,

- Event 2 tick 2 $x_2 = x_1$, $\Delta x = 0$.

$$\Delta t'_B = A\Delta t_B.$$

We have

$$\begin{aligned} D\Delta x &= AD\Delta x' + vAD\Delta t', \\ A\Delta t &= A^2\Delta t' + AD\Delta x'. \end{aligned}$$

Subtract

$$\begin{aligned} A\Delta t - D\Delta x &= A^2\Delta t' + AD\Delta x' - (AD\Delta x' + vAD\Delta t') \\ &= A^2\Delta t' - vAD\Delta t'. \end{aligned}$$

Solve for $\Delta t'_B$,

$$\begin{aligned} \Delta t'_B &= \frac{A\Delta t - D\Delta x}{A\left(A - vD\right)} \\ &= \frac{A\Delta t_B}{A\left(A - vD\right)} \\ &= \frac{\Delta t_B}{A - vD}. \end{aligned}$$

So for clock A and B,

$$\begin{aligned} \Delta t_A &= A\Delta t'_A, \\ \Delta t_B &= \frac{1}{A - vD}\Delta t'_B. \end{aligned}$$

with $h = k$,

$$\begin{aligned} \Delta t &= h\Delta t', \\ \Delta t' &= k\Delta t. \end{aligned}$$

Use

$$\Delta t_B = A\Delta t'_B.$$

We have

$$\begin{aligned} A &= \frac{1}{A - vD}, \\ A - vD &= \frac{1}{A}, \\ vD &= A - \frac{1}{A}, \\ D &= \frac{A - \frac{1}{A}}{v} \\ &= \frac{A}{v}\left(1 - \frac{1}{A^2}\right). \end{aligned}$$

So we get

$$\begin{aligned} \Delta x &= A\Delta x' + vA\Delta t', \\ \Delta t &= A\Delta t' + \frac{A}{v}\left(1 - \frac{1}{A^2}\right)\Delta x'. \end{aligned}$$

6.2.4 Light speed is same in each reference frame

So for light,

$$\Delta x' = c\Delta t',$$
$$\Delta x = c\Delta t.$$

We have the transformation equations for Δx and Δt as follows.

$$\Delta x = A\Delta x' + vA\Delta t',$$
$$\Delta t = A\Delta t' + \frac{A}{v}\left(1 - \frac{1}{A^2}\right)\Delta x'.$$

Divided again,

$$\begin{aligned}\frac{\Delta x}{\Delta t} &= \frac{Ac\Delta t' + vA\Delta t'}{A\Delta t' + \frac{A}{v}(1 - \frac{1}{A^2})c\Delta t'} \\ &= \frac{A\left(c + v\right)\Delta t'}{A\left(1 + \frac{c}{v}(1 - \frac{1}{A^2})\right)\Delta t'} \\ &= \frac{c + v}{1 + \frac{c}{v}\left(1 - \frac{1}{A^2}\right)} \\ &= c.\end{aligned}$$

Then

$$\begin{aligned}c + v &= c\left[1 + \frac{c}{v}(1 - \frac{1}{A^2})\right], \\ \frac{c + v}{c} &= 1 + \frac{c}{v}\left(1 - \frac{1}{A^2}\right), \\ 1 + \frac{v}{c} &= 1 + \frac{c}{v}\left(1 - \frac{1}{A^2}\right), \\ \frac{v}{c} &= \frac{c}{v}\left(1 - \frac{1}{A^2}\right), \\ 1 - \frac{1}{A^2} &= \frac{v^2}{c^2}, \\ A^2 &= \frac{1}{1 - \frac{v^2}{c^2}}.\end{aligned}$$

So that we have

$$A = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} = \gamma.$$

A is the Lorentz factor, we get

$$\Delta x = \gamma\left(\Delta x' + v\Delta t'\right),$$
$$\Delta t = \gamma\left(\Delta t' + \frac{1}{v}(1 - \frac{1}{A^2})\Delta x'\right).$$

Then determine $\frac{1}{v}(1 - \frac{1}{A^2})$ in terms of v and c.

$$\begin{aligned}\frac{1}{v}\left(1 - \frac{1}{A^2}\right) &= \frac{1}{v}\left(1 - \left(1 - \frac{v^2}{c^2}\right)\right) \\ &= \frac{v}{c^2}.\end{aligned}$$

Then we get all the components of the Lorentz transformation.

$$\Delta x = \gamma \left(\Delta x' + v \Delta t' \right),$$
$$\Delta t = \gamma \left(\Delta t' + \frac{v}{c^2} \Delta x' \right).$$

6.3 Leading Clock Lags / Trailing Clock Ahead

Let S' move rightward at speed v with respect to S, and let two events occur simultaneously in S at different locations,

$$\Delta t = 0,$$
$$\Delta x \neq 0.$$

From the inverse transformation

$$t' = \gamma \left(t - \frac{vx}{c^2} \right),$$

we obtain

$$\Delta t' = -\gamma \frac{v \Delta x}{c^2}.$$

If

$$\Delta x > 0,$$

then

$$\Delta t' < 0.$$

So in S' the leading clock (at larger x) has a smaller time reading, i.e. the leading clock lags.

6.4 Time dilation

Consider a clock at rest in S', so that $\Delta x' = 0$.
According to the Lorentz transformation,

$$\Delta x = \gamma \left(\Delta x' + v \Delta t' \right) = \gamma v \Delta t',$$
$$\Delta t = \gamma \left(\Delta t' + \frac{v}{c^2} \Delta x' \right) = \gamma \Delta t'.$$

This shows that the time interval Δt measured in S is longer than the proper time interval $\Delta t'$ measured in S'.

6.5 Length contraction

Consider a rod at rest in S' with length $\Delta x'$. According to the Lorentz transformation,

$$\Delta x = \gamma \left(\Delta x' + v \Delta t' \right),$$
$$\Delta t = \gamma \left(\Delta t' + \frac{v}{c^2} \Delta x' \right).$$

To measure the length of the rod in S, we need to consider the positions of the ends of the rod simultaneously in S, so $\Delta t = 0$.
Therefore

$$0 = \gamma\left(\Delta t' + \frac{v}{c^2}\Delta x'\right),$$
$$\Delta t' = -\frac{v}{c^2}\Delta x'.$$

Substituting this into the expression for Δx, we get

$$\begin{aligned}\Delta x &= \gamma\left(\Delta x' + v\left(-\frac{v}{c^2}\Delta x'\right)\right)\\ &= \gamma\left(1 - \frac{v^2}{c^2}\right)\Delta x'\\ &= \frac{\Delta x'}{\gamma}.\end{aligned}$$

This shows that the length of the rod measured in S is shorter than its proper length measured in S'.

7 Lecture 7: Space-time and Minkowski Diagrams

7.1 Velocity addition

A person moving with velocity v in a train throws a ball with velocity u relative to the train. The velocity of the ball relative to the ground u' is given by the relativistic velocity addition formula. In classical mechanics, we would simply add the velocities,

$$u' = u + v.$$

If velocity of ball and train are equal to $0.8c$, then according to classical mechanics,

$$u' = 0.8c + 0.8c = 1.6c,$$

which exceeds the speed of light, so we need to build a new relation. For an object moving with velocity u in frame S, and the frame S' moving with velocity v relative to S.

- Event 1 in S: $(0, 0)$.
- Event 2 in S: (x, t).
- Event 1 in S': $(0, 0)$.
- Event 2 in S': (x', t').
- $u = \frac{\Delta x}{\Delta t}$.
- $u' = \frac{\Delta x'}{\Delta t'}$.

Use the Lorentz transformation.

$$\Delta x' = \gamma(\Delta x - v\Delta t),$$
$$\Delta t' = \gamma\left(\Delta t - \frac{v}{c^2}\Delta x\right).$$

The velocity of the object in S' is given by

$$\begin{aligned} u' = \frac{\Delta x'}{\Delta t'} &= \frac{\gamma(\Delta x - v\Delta t)}{\gamma\left(\Delta t - \frac{v}{c^2}\Delta x\right)} \\ &= \frac{\Delta x - v\Delta t}{\Delta t - \frac{v}{c^2}\Delta x} \\ &= \frac{u - v}{1 - \frac{uv}{c^2}}, \end{aligned}$$

which is the relativistic velocity addition formula.

Notation

Then emphasize the importance of understanding the physical meaning of each symbol before applying the relativistic velocity addition formula.

- u velocity of the object measured in the S (Ground) frame.
- v velocity of the S' (Moving frame) relative to the S (Ground) frame.
- u' velocity of the object measured in the S' (Moving) frame.

Example

Luke sees Vader move to the right at $0.8c$, Luke sees Han move to the left at $0.7c$, how fast does Vader see Han move?

Using the relativistic velocity addition formula

$$u' = \frac{u - v}{1 - \frac{uv}{c^2}},$$

$u = 0.7c$ (Han's velocity to the right relative to Vader) and $v = -0.8c$ (Luke's frame relative to Vader's frame), so

$$u' = \frac{0.7c - (-0.8c)}{1 - \frac{(0.7c)(-0.8c)}{c^2}} = \frac{1.5c}{1 + 0.56} = \frac{1.5c}{1.56} \approx 0.962c.$$

Therefore, Vader sees Han move to the right at approximately $0.962c$.

Example

Spaceship A moves at $0.5c$ to the right approaching the earth, and Spaceship B moves at $0.5c$ to the left approaching the earth. How fast does Spaceship A see Spaceship B move?

Using the relativistic velocity addition formula

$$u' = \frac{u - v}{1 - \frac{uv}{c^2}},$$

$u = -0.5c$ (Spaceship B's velocity to the left) and $v = 0.5c$ (Spaceship A's velocity to the right), so

$$u' = \frac{-0.5c - 0.5c}{1 - \frac{(-0.5c)(0.5c)}{c^2}} = \frac{-1c}{1 + 0.25} = \frac{-1c}{1.25} \approx -0.8c.$$

Therefore, Spaceship A sees Spaceship B move to the left at approximately $0.8c$.

7.2 Summary of Lorentz Transformation

7.2.1 Lorentz Transformation

The Lorentz transformation relates the space and time coordinates of events as observed in two inertial frames moving at a constant velocity relative to each other. If S and S' are two such frames, with S' moving at velocity v relative to S along the x-axis, the transformations are

$$\begin{aligned} x' &= \gamma(x - vt), \\ t' &= \gamma\left(t - \frac{vx}{c^2}\right), \\ y' &= y, \\ z' &= z. \end{aligned}$$

where the Lorentz factor is given by

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}.$$

These transformations ensure that the speed of light is constant in all inertial frames and lead to phenomena such as time dilation and length contraction.

7.2.2 Low-speed Approximation

At speeds much less than the speed of light ($v \ll c$), the Lorentz factor γ approaches 1, and the Lorentz transformations reduce to the classical Galilean transformations.

$$\begin{aligned} x' &\approx x - vt, \\ t' &\approx t, \\ y' &= y, \\ z' &= z. \end{aligned}$$

to be precise we should also require $vx/c^2 \ll 1$ for the approximation $t' \approx t$ to hold.

7.3 The Invariant Spacetime Interval

Definition 7.1

The spacetime interval (or invariant interval) between two events is a quantity that remains unchanged under Lorentz transformations. It is defined as

$$s^2 = c^2\Delta t^2 - \Delta x^2,$$

where Δt, Δx are the differences in the time and spatial coordinates of the events.

Using the Lorentz transformations, we can show that this interval remains the same in all inertial frames.

$$x = \gamma(x' + vt),$$
$$t = \gamma\left(t' + \frac{vx'}{c^2}\right).$$

Substituting the Lorentz transformations into the spacetime interval, we get

$$\begin{aligned}
c^2\Delta t^2 - \Delta x^2 &= c^2\left(\gamma\left(t' + \frac{vx'}{c^2}\right)\right)^2 - (\gamma(x' + vt'))^2 \\
&= \gamma^2\left[c^2\left(t' + \frac{vx'}{c^2}\right)^2 - (x' + vt)^2\right] \\
&= \gamma^2\left[c^2t'^2 + 2c^2t'\frac{vx'}{c^2} + c^2\left(\frac{vx'}{c^2}\right)^2 - x'^2 - 2x'vt - v^2t^2\right] \\
&= \gamma^2\left[c^2t'^2 + 2vt'x' + \frac{v^2x'^2}{c^2} - x'^2 - 2vt'x' - v^2t^2\right] \\
&= \gamma^2\left[c^2t'^2 - v^2t^2 - x'^2 + \frac{v^2x'^2}{c^2}\right] \\
&= \gamma^2\left[c^2t'^2 - v^2t^2 - x'^2\left(1 - \frac{v^2}{c^2}\right)\right] \\
&= \gamma^2\left[c^2t'^2 - v^2t^2 - x'^2\frac{c^2 - v^2}{c^2}\right] \\
&= \gamma^2\left[c^2t'^2 - v^2t^2 - \frac{c^2 - v^2}{c^2}x'^2\right] \\
&= c^2t'^2 - x'^2.
\end{aligned}$$

We actually need to calculate

$$\Delta s^2 = c^2\Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2.$$

Example: Time dilation

Consider frame S' moving with velocity v relative to frame S along the x-axis. Let a clock be at rest in S' at position $x' = 0$. The time interval between two ticks of this clock in S' is $\Delta t'$. The invariant spacetime interval gives

$$c^2\Delta t^2 - \Delta x^2 = c^2\Delta t'^2 - \Delta x'^2.$$

Since the clock is at rest in S', $\Delta x' = 0$, so

$$c^2\Delta t^2 - \Delta x^2 = c^2\Delta t'^2.$$

In frame S, the clock moves with velocity v, so $\Delta x = v\Delta t$. Substituting this, we get

$$c^2\Delta t^2 - (v\Delta t)^2 = c^2\Delta t'^2,$$

$$\Delta t^2(c^2 - v^2) = c^2\Delta t'^2,$$

$$\Delta t^2 = \frac{c^2\Delta t'^2}{c^2 - v^2} = \frac{\Delta t'^2}{1 - \frac{v^2}{c^2}},$$

$$\Delta t = \frac{\Delta t'}{\sqrt{1 - \frac{v^2}{c^2}}} = \gamma\Delta t'.$$

7.4 Minkowski Diagram

7.4.1 Minkowski Diagram Basics

Definition 7.2

A Minkowski diagram is a two-dimensional graphical representation of spacetime, where the horizontal axis represents space and the vertical axis represents time.
It is used to visualize the relationships between different inertial frames and the effects of special relativity, such as time dilation and length contraction.

Difference with respect to classical diagram

1. distance x displayed on the horizontal axis.
2. time ct displayed on the vertical axis.
3. units of space and time chosen so that an object moving with a speed of light c is depicted as traveling at 45 degrees line in the diagram.

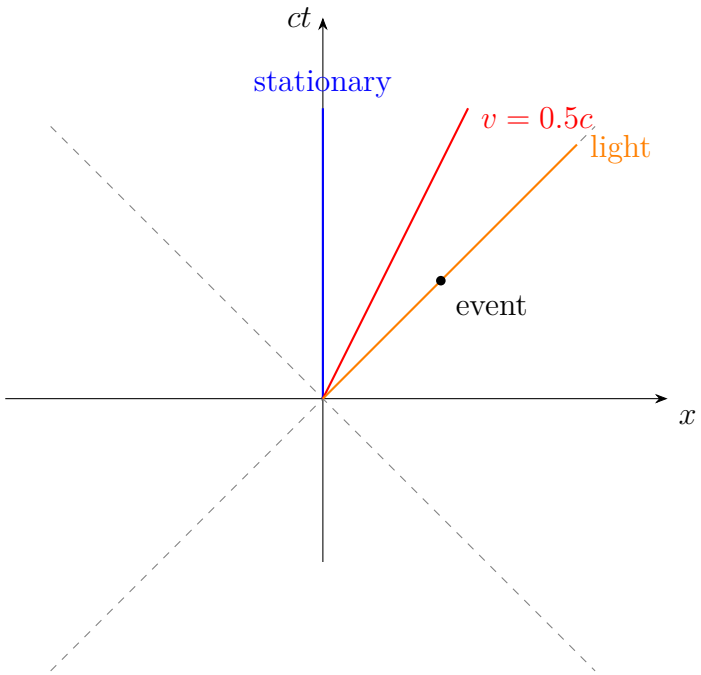


Figure 1: Basic Minkowski diagram: x horizontal, ct vertical, light worldline at 45°.

7.4.2 Worldlines

Definition 7.3

A worldline represents the path of an object through spacetime.

- For stationary objects, their worldlines are vertical lines in the diagram.
- For objects moving with constant velocity, their worldlines are straight lines with a slope determined by their velocity; the steeper the slope, the slower the object moves in space.

Remark

In the ct–x plane, if slope is defined as

$$k = \frac{dy}{dx} = \frac{d(ct)}{dx},$$

then the light worldline has slope 1 (45°). Any material object has a steeper worldline

$$\left|\frac{d(ct)}{dx}\right| > 1,$$

must lie inside the light cone.

7.4.3 Event Simultaneity

Event simultaneity is represented by horizontal lines in the diagram.

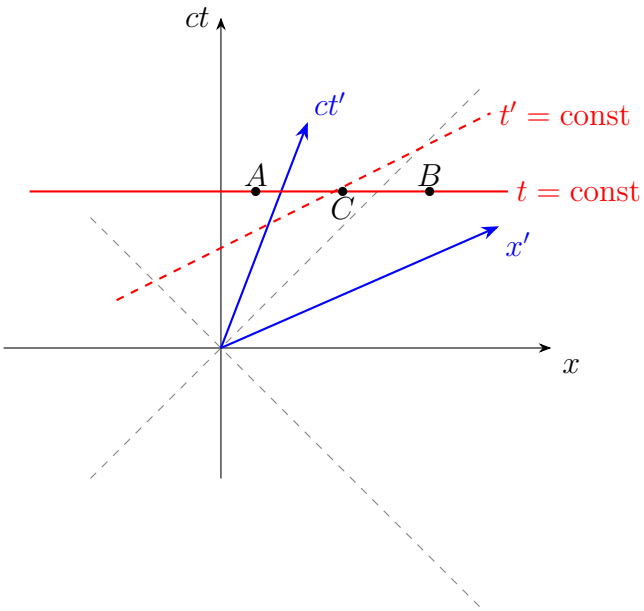


Figure 2: Relativity of simultaneity: lines of constant t are horizontal, while lines of constant t' are tilted.

7.4.4 Time Dilation

Time dilation is represented by the difference in the slopes of the worldlines of moving clocks compared to stationary clocks. A moving clock's worldline is less steep than that of a stationary clock, indicating that less time elapses for the moving clock between two events.

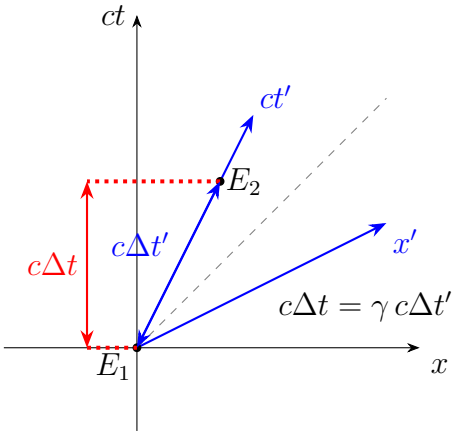


Figure 3: Time dilation: E_2 lies on the ct' axis; the vertical time interval in S is γ times the proper time in S'.

> **Remark**
>
> To read γ from the diagram, we need to find the intersection of the ct' axis with a line of constant ct, which gives $ct = \gamma ct'$.

7.4.5 Length Contraction

Length contraction is represented by the apparent shortening of the spatial distance between two points along the direction of motion for a moving object. In the diagram, the spatial separation between the endpoints of a moving object is smaller than that of a stationary object, illustrating that lengths contract along the direction of motion.

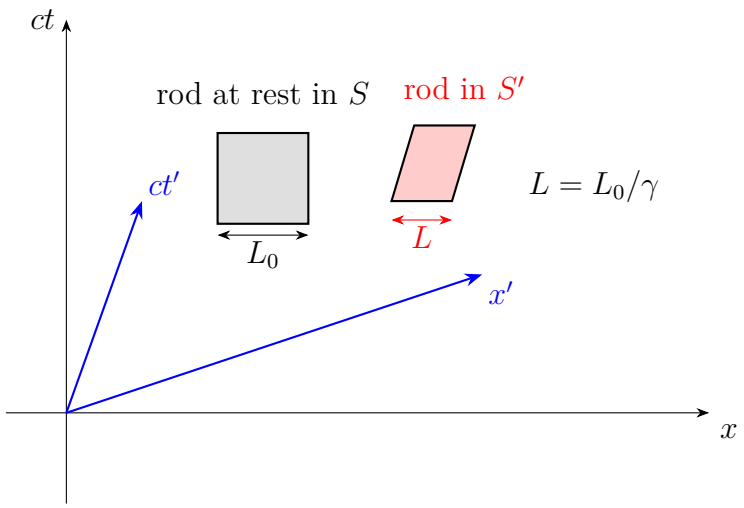


Figure 4: Length contraction: a moving rod appears shorter when measured simultaneously in S.

> **Remark**
>
> Simultaneity lines are parallel to the x' axis. Length contraction is read by intersecting the rod's worldlines with lines parallel to the x' axis.

7.4.6 Lorentz Transformations and Tilted Axes

The Lorentz transformations are

$$x' = \gamma(x - vt), \qquad t' = \gamma\left(t - \frac{vx}{c^2}\right), \qquad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}.$$

In a Minkowski diagram

- the ct' axis is the worldline

$$x' = 0 \implies x = vt, \text{i.e. } ct = (c/v)x, \text{ slope } c/v.$$

- the x' axis is the line of simultaneity

$$t' = 0 \implies ct = (v/c)x, \text{ slope } v/c.$$

- the light cone $x = \pm ct$ remains at 45° in both inertial frames.

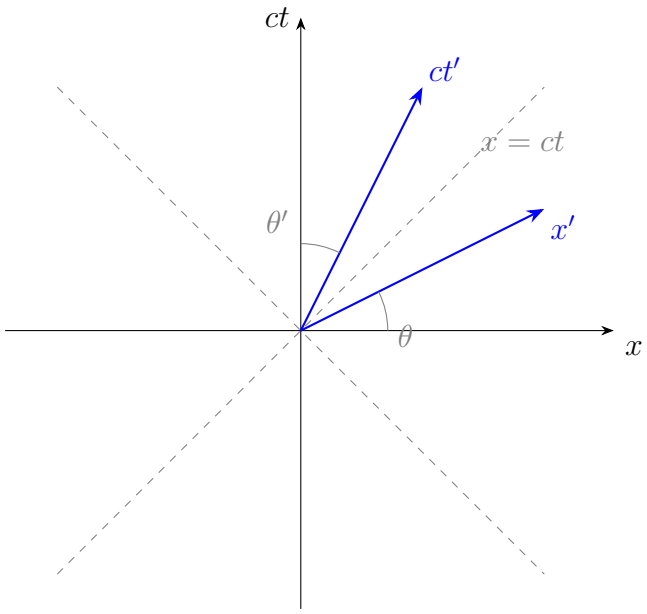


Figure 5: When S' moves with velocity v relative to S, the ct' and x' axes squeeze toward the light cone.

7.4.7 Light Cone and Causal Structure

The spacetime interval is invariant under Lorentz transformations.

$$s^2 = c^2t^2 - x^2.$$

- $s^2 > 0$: timelike separation; events can be causally connected.
- $s^2 = 0$: lightlike separation; connected by a light signal.
- $s^2 < 0$: spacelike separation; no causal connection.

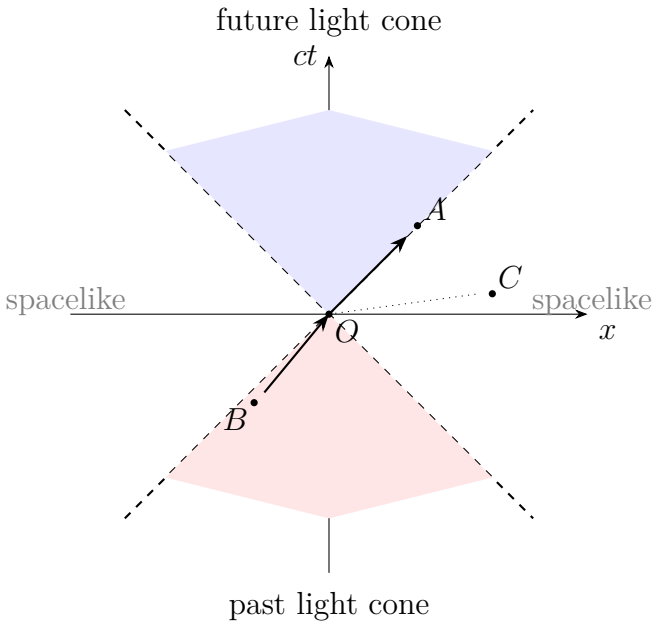


Figure 6: The light cone divides spacetime into future, past, and spacelike regions.

7.4.8 Invariant Spacetime Interval and Hyperbolae

For a fixed s^2, Lorentz transformations move events along hyperbolae.

$$c^2t^2 - x^2 = \pm a^2.$$

This is analogous to rotations in the x–y plane preserving $x^2 + y^2$.

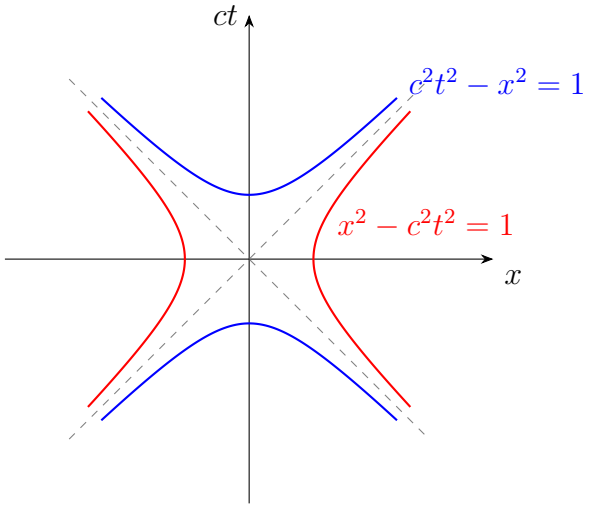


Figure 7: Hyperbolae of constant interval: Lorentz transformations slide events along these curves.

7.4.9 The Absolute Future and the Absolute Past

In a Minkowski diagram, the light cone divides spacetime into regions that are invariant under Lorentz transformations. The interior of the future light cone is called the *absolute future* of the event at the origin O, while the interior of the past light cone is called the *absolute past*.

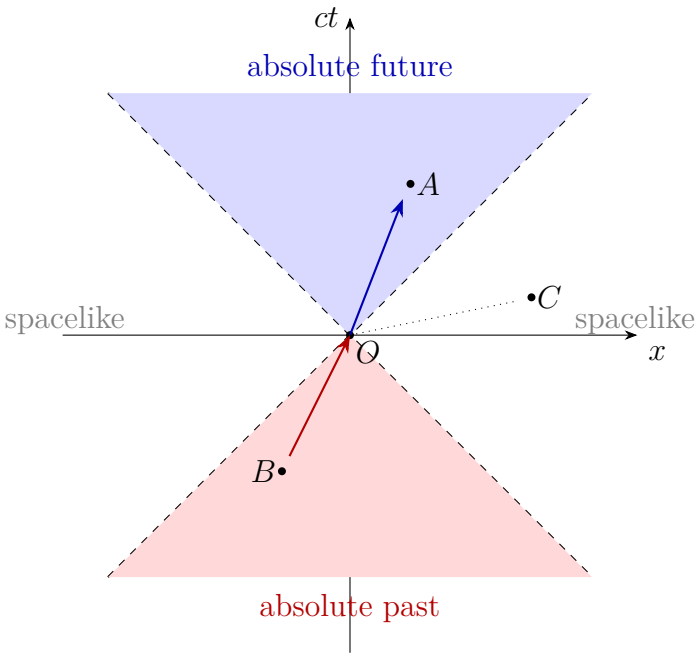


Figure 8: The absolute future and the absolute past: events inside the future light cone can be causally influenced by O, events inside the past light cone can causally influence O, and events outside the light cone are spacelike separated from O.

1. For any event A inside the future cone,

$$s^2 = c^2 t_A^2 - x_A^2 > 0, \quad t_A > 0.$$

2. For any event B inside the past cone,

$$s^2 = c^2 t_B^2 - x_B^2 > 0, \quad t_B < 0.$$

Because the sign of s^2 and the time ordering are preserved by Lorentz transformations, every inertial observer agrees that A occurs after O and that B occurs before O. And since A and B lie inside the light cone, a signal traveling slower than or at the speed of light can connect them to O.

$$O \prec A, \quad B \prec O.$$

Events outside the light cone are spacelike separated ($s^2 < 0$) and belong to neither the absolute future nor the absolute past.

8 Lecture 8: Relativistic Doppler Effect and Dynamics

8.1 Recall: Classical Doppler Effect

Definition 8.1

The classical Doppler effect describes the change in frequency (or wavelength) of a wave due to the relative motion of the source and the observer along the line of sight.

When the source is moving directly towards the observer with velocity v, the observed frequency f' is related to the emitted frequency f by

$$f' = \frac{f}{1 - v/c}.$$

8.2 Relativistic Doppler Effect

8.2.1 Definition

Definition 8.2

The relativistic Doppler effect describes the change in frequency (or wavelength) of light or other electromagnetic waves due to the relative motion of the source and the observer along the line of sight.

When the source is moving directly towards or away from the observer with velocity v, the observed frequency f' is related to the emitted frequency f by

$$f' = f\sqrt{\frac{1+\beta}{1-\beta}}.$$

And for transverse motion, the relativistic Doppler effect is

$$f' = \frac{f}{\gamma}.$$

Notation

Here, β is the ratio of the source's velocity to the speed of light.

$$\beta = \frac{v}{c}.$$

Example

A receding source appears redder (redshift); an approaching source appears bluer (blueshift).

Example

Green light (530 nm) appears blue (470 nm) when the source approaches. The required frequency increase is $530/470 = 1.128$, giving $\beta = -0.12$, i.e. $v = 12\%$ of c towards the observer.

8.2.2 Derivation

The relativistic Doppler effect can be derived using the Lorentz transformation. Consider a light wave emitted from a source moving with velocity v relative to an observer. Let the time interval between successive wave fronts in the source frame be Δt. In the observer frame, the time interval $\Delta t'$ is related by the time dilation formula

$$\Delta t' = \gamma\left(\Delta t - \frac{v\Delta x}{c^2}\right).$$

For light, $\Delta x = c\Delta t$, so

$$\Delta t' = \gamma\left(\Delta t - \frac{vc\Delta t}{c^2}\right) = \gamma\Delta t\left(1 - \frac{v}{c}\right).$$

Using $\gamma = \frac{1}{\sqrt{1-\beta^2}}$ and $\beta = \frac{v}{c}$, we get

$$\Delta t' = \Delta t\sqrt{\frac{1-\beta}{1+\beta}}.$$

Since frequency is the inverse of the time interval, $f' = \frac{1}{\Delta t'}$ and $f = \frac{1}{\Delta t}$, we obtain the relativistic Doppler formula

$$f' = f\sqrt{\frac{1+\beta}{1-\beta}}.$$

8.2.3 Cosmological Redshift

The modern understanding of cosmological redshift is based on the expansion of the universe, not the relative motion of celestial objects.

$$1 + z = \frac{\lambda_{\text{obs}}}{\lambda_{\text{em}}} = \frac{a(t_{\text{obs}})}{a(t_{\text{em}})},$$

where z is the redshift, λ_{obs} and λ_{em} are the observed and emitted wavelengths, and $a(t)$ is the scale factor of the universe at time t.

Example

A hydrogen atom emits the $n = 3 \to 2$ transition with $\lambda = 660$ nm. A star receding at $v = 0.8c$ has $\beta = 0.8$, so

$$\frac{f'}{f} = \sqrt{\frac{1-\beta}{1+\beta}} = \sqrt{\frac{0.2}{1.8}} = \frac{1}{3}.$$

The observed wavelength is $\lambda' = 3\lambda = 1980$ nm.

8.3 Introduction to Dynamics

8.3.1 Momentum and Energy Conservation in Classical Mechanics

Definition 8.3

In classical mechanics, momentum is conserved in isolated systems.

$$\vec{p}_{\text{tot}} = \sum_i \vec{p}_i = \text{const.}$$

derivation follows from Newton's second law applied to each particle in the system.

$$\frac{d\vec{p}_i}{dt} = \vec{F}_i,$$

$$\sum_i \frac{d\vec{p}_i}{dt} = \sum_i \vec{F}_i = 0 \implies \frac{d\vec{p}_{\text{tot}}}{dt} = 0.$$

For equal-mass elastic scattering, conservation laws imply the outgoing particles are perpendicular,

$$\vec{p}_1 \perp \vec{p}_2.$$

Definition 8.4

In classical mechanics, the total energy of an isolated system is conserved.

$$E_{\text{tot}} = \sum_i E_i = \text{const.}$$

This follows from the work-energy theorem applied to each particle in the system.

$$\frac{dE_i}{dt} = \vec{F}_i \cdot \vec{v}_i,$$

$$\sum_i \frac{dE_i}{dt} = \sum_i \vec{F}_i \cdot \vec{v}_i = 0 \implies \frac{dE_{\text{tot}}}{dt} = 0.$$

Remark

Experimental results from cloud chambers, bubble chambers, and the LHC show that

$$\vec{p} = m\vec{v}$$

is not conserved in relativistic scenarios.

9 Lecture 9: Relativistic Dynamics

9.1 Momentum in Relativistic Mechanics

9.1.1 Derivation of Relativistic Momentum

Consider a particle moving in orthogonal directions in frames S and S', with S' moving with speed v relative to S along the x-axis.

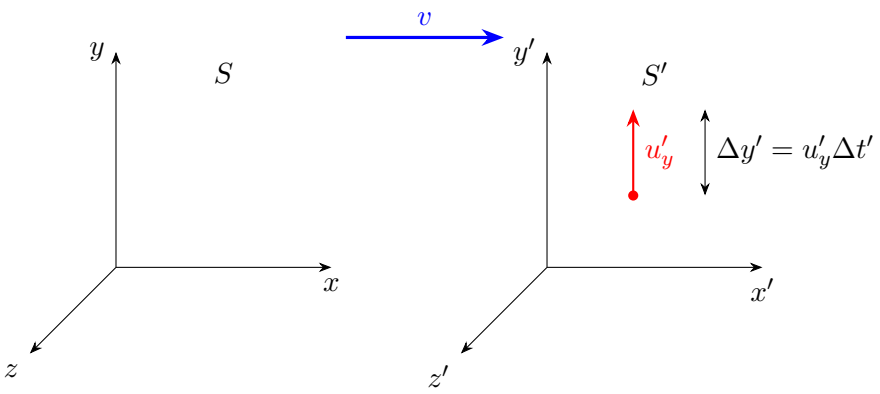


Figure 9: Standard configuration. Frame S' moves with velocity v along $+x$ relative to S. A particle moves vertically with speed u'_y in S'.

Transverse lengths are unchanged, $\Delta y = \Delta y'$, and time dilation gives $\Delta t = \gamma \Delta t'$. Hence

$$u_y = \frac{\Delta y}{\Delta t} = \frac{\Delta y'}{\gamma \Delta t'} = \frac{u'_y}{\gamma}.$$

Assume that relativistic momentum has the form

$$\vec{p} = km\vec{v},$$

with $k = k(v/c)$ to be determined.

9.1.2 Transverse Scattering in a Relativistic Frame

- Consider two identical particles A and B in a frame S' that moves with speed v close to c relative to S.
- In S', particle A moves rapidly along x' with a small transverse speed in y', while particle B moves slowly downward in the y' direction.
- After scattering, B moves slowly upward and A moves slowly downward, but with a large horizontal speed along x'.
- In frame S, the motion of B is relativistic.

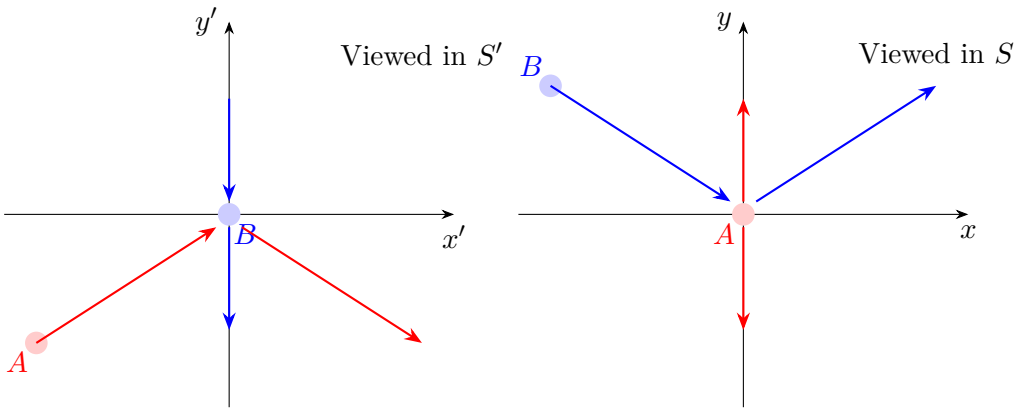


Figure 10: Transverse scattering of identical particles A and B. In S', B moves slowly and A is relativistic. In S, A moves slowly and B is relativistic.

By symmetry, the transverse speed of each particle in its own x rest frame is the same; call it U, with $U \ll c$. In S', the transverse speed of B is U. Since $u_y = u'_y/\gamma$, the transverse speed of B in S is

$$u_B^{(y)} = \frac{U}{\gamma}.$$

Particle B is relativistic in S, so

$$\Delta p_B^{(y)} = 2km\frac{U}{\gamma}.$$

Particle A is classical in S, with transverse speed U, so

$$\Delta p_A^{(y)} = 2mU.$$

Conservation of transverse momentum in S requires

$$2mU = 2km\frac{U}{\gamma},$$

hence

$$k = \gamma,$$

and therefore

$$\vec{p} = \gamma m\vec{v}.$$

Here v is the relativistic speed of the particle in frame S.

9.1.3 Definition of Relativistic Momentum

Definition 9.1

In relativistic mechanics, momentum is given by

$$\vec{p} = \gamma m\vec{v},$$

where

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

is the Lorentz factor, m is the rest mass of the particle, and $\vec{v}$ is its velocity in the given frame.

The total momentum of an isolated system is still conserved,

$$\vec{p}_{\text{tot}} = \sum_i \vec{p}_i = \text{const}.$$

Remark

At high speed the particle's inertia increases by the Lorentz factor γ. The rest mass m is invariant. The expression $\vec{p} = \gamma m\vec{v}$ is sometimes described as if the mass increases, but modern usage keeps m as the rest mass.

9.2 Relativistic Energy

Consider motion along one dimension $\hat{x}$. The force is

$$\vec{F} = \frac{d\vec{p}}{dt} = \frac{d(\gamma m\vec{v})}{dt}.$$

The work done by the force changes the kinetic energy T:

$$\begin{aligned} dT &= \vec{F} \cdot d\vec{x} \\ &= \frac{d(\gamma m\vec{v})}{dt} \cdot d\vec{x} \\ &= m\frac{d(\gamma v)}{dt}\, dx. \end{aligned}$$

Using $dx/dt = v$,

$$dT = m\left(\gamma v\, dv + v^2\, d\gamma\right).$$

From

$$\gamma^2\left(1 - \frac{v^2}{c^2}\right) = 1$$

we obtain

$$\gamma^2 c^2 = \gamma^2 v^2 + c^2.$$

Differentiating with respect to γ,

$$2\gamma c^2 = 2\gamma^2 v\frac{dv}{d\gamma} + 2\gamma v^2,$$

or

$$c^2\, d\gamma = \gamma v\, dv + v^2\, d\gamma.$$

Therefore

$$dT = mc^2\, d\gamma.$$

Integrating

$$\int dT = \int mc^2\, d\gamma,$$

We have

$$T = mc^2\gamma + A.$$

Since $T = 0$ when $v = 0$ and $\gamma = 1$, we have

$$A = -mc^2,$$

so

$$T = (\gamma - 1)mc^2.$$

The total energy is

$$E = \gamma mc^2 = mc^2 + T.$$

Here mc^2 is the rest energy.

Definition 9.2

The relativistic kinetic energy is

$$T = (\gamma - 1)mc^2.$$

The total energy is

$$E = \gamma mc^2 = mc^2 + T,$$

where mc^2 is the rest energy.

In the low-speed limit $v \ll c$,

$$\gamma \approx 1 + \frac{v^2}{2c^2},$$

so

$$T \approx \frac{1}{2}mv^2.$$

9.3 Invariant quantity

The quantity $E^2 - (pc)^2$ is invariant under Lorentz transformations, where E is the total energy and p is the relativistic momentum.

$$\begin{aligned} E^2 - (pc)^2 &= \gamma^2 m^2 c^4 - (\gamma mvc)^2 \\ &= \gamma^2 m^2 c^4 \left(1 - \frac{v^2}{c^2}\right) \\ &= \gamma^2 m^2 c^2 (c^2 - v^2) \\ &= m^2 c^4. \end{aligned}$$

Hence

$$E^2 = (pc)^2 + (mc^2)^2.$$

Definition 9.3

The relation between energy and momentum is

$$E^2 - (pc)^2 = m^2 c^4.$$

This relation holds for all particles, regardless of their velocity, invariant under Lorentz transformations.

Remark

For a particle at rest, $p = 0$, so

$$E = mc^2.$$

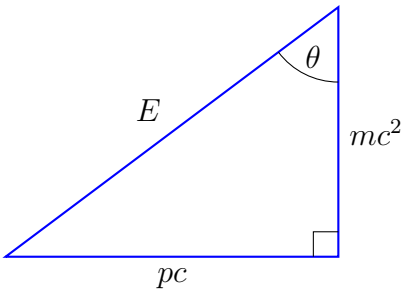
For a photon, $m = 0$, so

$$E = pc,$$

And

$$p = \frac{hf}{c}.$$

And the relation between energy and momentum is a right triangle, as shown in Figure below.



$\sin\theta = pc/E = v/c$

Figure 11: The energy–momentum relation as a right triangle

In this right triangle,

$$E^2 = (pc)^2 + (mc^2)^2.$$

The angle satisfies

$$\sin\theta = pc/E = v/c.$$

9.4 Summary

The summary on energy and momentum is:

- Momentum is
$$\vec{p} = \gamma m\vec{v}.$$
- Kinetic energy is
$$T = (\gamma - 1)mc^2.$$
- Total energy is
$$E = \gamma mc^2 = mc^2 + T.$$
- The energy–momentum invariant is
$$E^2 - (pc)^2.$$
- The energy–momentum relation is
$$E^2 = (pc)^2 + (mc^2)^2.$$
- Momentum is conserved.
- Energy is conserved.

9.5 Natural Units

With c retained,

- Energy
E is measured in MeV.

- Mass

$$m \text{ is measured in MeV}\, c^{-2}.$$

- Momentum

$$p \text{ is measured in MeV}\, c^{-1}.$$

Example

For a proton, $m_p = 938.3$ MeV c^{-2} and $T = 2000$ MeV. Then

$$E = T + m_p c^2 = 2938.3 \text{ MeV}, \qquad \gamma = \frac{E}{m_p c^2} = 3.132,$$

$$p = \frac{(E^2 - m_p^2 c^4)^{1/2}}{c} = 2784 \text{ MeV}\, c^{-1}.$$

Example

For an electron with $E = 1$ MeV and $m_e c^2 = 0.511$ MeV,

$$\gamma = \frac{1}{0.511} = 1.96, \qquad \frac{v}{c} = (1 - \gamma^{-2})^{1/2} = 0.86.$$

Thus $v = 0.86c$.

Example

Complete conversion of 1 kg of mass would release

$$E = mc^2 \approx 9 \times 10^{16} \text{ J} \approx 10^{17} \text{ J},$$

about the annual electricity consumption of 3×10^6 households, or about 21 kT of TNT.

9.6 Analogies between invariants

The spacetime interval and the energy–momentum invariant play analogous roles. The correspondence is

$$E \leftrightarrow ct, \qquad pc \leftrightarrow x.$$

The energy–momentum four-vector is

$$\left(\frac{E}{c}, p_x, p_y, p_z\right),$$

Which is analogous to

$$(ct, x, y, z)\,.$$

Under a Lorentz boost along x, the energy–momentum transformations and the spacetime coordinate transformations have the same form

$$\begin{cases} E' = \gamma(E - v p_x), \\ p'_x = \gamma\left(p_x - \dfrac{vE}{c^2}\right), \\ p'_y = p_y, \\ p'_z = p_z, \end{cases} \qquad \begin{cases} x' = \gamma(x - vt), \\ t' = \gamma\left(t - \dfrac{vx}{c^2}\right), \\ y' = y, \\ z' = z. \end{cases}$$

The invariant is

$$E^2 - (pc)^2 = (E')^2 - (p'c)^2 = m^2c^4,$$

just as

$$c^2t^2 - x^2 = (ct')^2 - (x')^2.$$

Thus, in the same way as for length and time, one obtains the Lorentz transformations for energy and momentum.

9.7 Example From Problem Sheet

Example

1. γ_A, v_A?

$$T = (\gamma_A - 1)mc^2.$$

Solving for γ_A,

$$\gamma_A = \frac{T}{mc^2} + 1 = 1.155.$$

Then the velocity v_A can be found from γ_A,

$$v_A = c\sqrt{1 - \frac{1}{\gamma_A^2}} = 0.514c.$$

2.

9.8 Further Reading

A Simplified Derivations

A.1 Lorentz Transformation

A.1.1 Derivation of Lorentz Transformation: Four Facts

We use four facts.

1. S' moves with velocity v with respect to S.
2. S moves with velocity $-v$ with respect to S'.
3. The laws of physics are the same in both frames, so time dilation looks the same from either frame.
4. A light pulse has speed c in both S and S'.

Start with linear transformations

$$x = Ax' + Bt', \qquad t = Ct' + Dx'.$$

Fact 1

Consider a point fixed in S' ($x' = 0$). It moves with velocity v in S,

$$x = Bt', \quad t = Ct' \implies \frac{x}{t} = \frac{B}{C} = v \implies B = vC.$$

Fact 2

Considering a point fixed in S ($x = 0$) moves with velocity $-v$ in S',

$$0 = Ax' + Bt' \implies \frac{x'}{t'} = -\frac{B}{A} = -v \implies A = C.$$

Thus

$$x = A(x' + vt'), \qquad t = At' + Dx'.$$

Fact 3

We use the fact that time dilation is symmetric. A clock at rest in S' ($x' = 0$) gives $t = At'$. A clock at rest in S ($x = 0$) gives $t' = t/(A - vD)$.

$$A = \frac{1}{A - vD} \implies D = \frac{1}{v}\left(A - \frac{1}{A}\right).$$

Fact 4

And lastly, the light speed invariant. If $x' = ct'$, then $x = ct$.

$$c = \frac{x}{t} = \frac{A(ct' + vt')}{At' + \frac{1}{v}\left(A - \frac{1}{A}\right)ct'}.$$

Solving gives

$$A = \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}.$$

Then

$$x = \gamma(x' + vt'), \qquad t = \gamma\left(t' + \frac{v}{c^2}x'\right).$$

Inverting

$$x' = \gamma(x - vt), \qquad t' = \gamma\left(t - \frac{v}{c^2}x\right).$$

A.1.2 Using Linearity and Postulates

Instead of using four undetermined coefficients (A, B, C, D), we use linear assumptions and the two postulates of Special Relativity to derive the transformation quickly and avoid algebraic mess.

Step 1: Linearity Assumption

Assume the transformation is linear. Let frame S' move with velocity v along the positive x-axis relative to frame S. We propose

$$x' = \gamma(x - vt), \tag{1}$$
$$t' = \gamma(t - \alpha x), \tag{2}$$

where γ and α are constants to be determined.

Step 2: Invariance of the Speed of Light (View from S)

Consider a light pulse emitted at the origin at $t = 0$. In frame S, its position is $x = ct$. In frame S', the speed of light must also be c, so $x' = ct'$. Substitute $x = ct$ and $x' = ct'$ into equation (1),

$$ct' = \gamma(ct - vt) = \gamma(c - v)t. \tag{3}$$

Substitute $x = ct$ into equation (2).

$$t' = \gamma(t - \alpha ct) = \gamma(1 - \alpha c)t. \tag{4}$$

Divide equation (3) by equation (4).

$$\frac{ct'}{t'} = c = \frac{\gamma(c - v)t}{\gamma(1 - \alpha c)t} \implies c = \frac{c - v}{1 - \alpha c}. \tag{5}$$

Solving for α.

$$c(1 - \alpha c) = c - v \implies c - \alpha c^2 = c - v \implies \alpha = \frac{v}{c^2}. \tag{6}$$

Step 3: Principle of Relativity (View from S')

From the perspective of frame S', frame S moves with velocity $-v$. By the principle of relativity, the transformation from S' to S must have the same form, with v replaced by $-v$. Substitute $x' = ct'$ and $x = ct$ into the inverse transformation (or use the symmetry of the relative velocity).

$$x = \gamma(x' + vt') \implies ct = \gamma(ct' + vt'). \tag{7}$$

Using the known relation from Step 2, and requiring the transformations to be consistent, we solve for γ.

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}. \tag{8}$$

Step 4: Conclusion

Substituting α and γ back into the initial equations,

$$\begin{aligned} x' &= \gamma(x - vt), \\ t' &= \gamma\left(t - \frac{v}{c^2}x\right). \end{aligned}$$